Assignment 2.

Name: M. Usman Saleem

Topic: Advanced IP Subnetting

Course: NIAI Cyber Security

Part 1: Class C Subnetting Scenarios.

1: The Small Law Firm:

192.168.15.0/24

192.168.15.0/24

Decimal mask:

11111111.11111111.11111111.00000000

Required subnets = 4

$2^2 = 4$

Borrow 2 bits: 24+2=26

192.168.15.0/26

Decimal mask:

11111111.11111111.11111111.11000000

255.255.255.192

CamScanner

Hosts Per Subnet:

Total host bits: $8-2=6$

2^6-2

$64-2=62$

Useable Host $=62$

Block Size:

$256-192=64$

2nd Subnet:

Network id: 192.168.15.64

Range: 192.168.15.65 — 192.168.15.126

Broadcast id: 192.168.15.127.

Scenario 2: The Medical Clinic:

192.168.100.0/24.

Decimal mask:

11111111.11111111.11111111.00000000

Required subnets: 6

$2^3=8$

Borrow 3 bits: $24+3=27$

192.168.100.0/27

Decimal mask:

11111111.11111111.11111111.11100000

255.255.255.224

Block Size:

256 − 224 = 32

Subnet increment = 32

First Subnet: 200.50.25.0 (NI)

Broadcast Id: 200.50.25.31

Usable: 200.50.25.1 — 200.50.25.30

Scenario 3: High School

192.168.100.0/24

Decimal mask:

11111111. 11111111. 11111111. 00000000

4 subnets

Each 250 hots.

2^6 = 6 bits needed.

Totats bits = 32

32 − 6 = 26

Subnet mask:

CIDR: /26

Decimal:

255.255.255.192

Block Size:

$256 - 192 = 64$

Subnet increment = 64

4th Subnet Range: 192.168.100.193 — 192.168.100.254

Scenario 4- Coffee Shop Chain:

198.100.50.0/24

Requirement 14 hosts per subnet.

Host Bits:

$2^4 - 2 = 14$

Host bits = 4

Find Prefix: $32 - 4 = 28$

Subnet mask = 255.255.255.240

Block Size: $256 - 240 = 16$

Subnet increased by 16

Subnet List:

198.100.50.0
.16
.32
.48
.64
.80
.96
.112
.128
.176
.192
.208
.224
.240

Total = 16 Subnets

3rd Subnet:

Network ID: 198.100.50.32

Scenario 5: E-Commerce Startup:

Network: 203.0.113.0/24

Need 12 subnets.

Borrows bits

$2^n \geq 12$

$2^n \geq 12$

$2^4 \geq 12$

4 bits Borrow

New Mask: 24+4 = 28

11111111.11111111.11111111.11110000

Decimal:

255.255.255.240

Host bits = 8 - 4 = 4

Hosts:

$2^4 - 2 = 14$

Block size = 256 - 240 = 16

CamScanner

Last Subnet:

Network Id:

203.0.113.240

Range = 203.0.113.241 – 203.0.113.254

Broadcast: 203.0.113.255.

Part 2: Class B subnetting Scenarios.

Scenario 6: The Regional Hospital

172.16.0.0/16

Need subset 50.

$2^6 = 64$

Borrow 6 bits.

16+6 = 22

172.16.0.0/22

11111111.11111111.111111 00.00000000

Subnet mask

255.255.252.0

Host Bits:

$32 - 22 = 10$

$2^{10} - 2 = 1022$

Block Size:

$256 - 252 = 4$

2nd Subnet:

Network ID: 172.16.4.0

Broadcast: 172.16.7.255

Scenario 7 - Corporate HQ:

Need 500 hosts

$2^9 = 510$

Host bits = 9

Prefix = 32 - 9 = 23

Subnet Mask:

255.255.254.0

Block Size = 256 - 254 = 2

Block Increment = 2.

CamScanner

Scenario 8 - Smart City:

Need 100 Subnets.

$2^7 = 128$

Borrow 7 bits

prefix = 16 + 7 = 23

Subnet mask = 255 255 254.0

Hosts = $2^9 - 2 = 510$

3rd Subnet:

Block size = 2

Subnets:

0
2
4

3rd Subnet: 128.10.4.0.

CS CamScanner

Scenario 9 – University Wi-Fi

Need 2000 hosts

Host bits =

$2^{11} - 2 = 2046$

Host bits = 11

Prefix : 32 − 11 = 21

Mask = 255.255.248.0

Block size = 256 − 248 = 8

First Subnet: 172.31.0.0

Range : 172.31.0.1 — 172.31.7.254

Broadcast: 172.31.7.254

CS CamScanner

Scenario 10 - Manufacturing Plant

Network : 150.20.0.0/16

need 16 subnets.

Borrow Bits: $2^4 = 16$

Borrow 4 bits.

New Prefix: 16+4 = 20

Subnet mask = 255.255.240.0

Hosts: 32 − 20 = 12

Hosts = $2^{12} - 2 = 4094$

Block size = 256 − 240 = 16

5th Subnet:

0
16
32
48
64

Network : 150.20.64.0

Range = 150.20.64.1 − 150.20.79.254

Broadcast: 150.20.79.255

CamScanner